Pritish Saha

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Education

Indian Institute of Technology, Kharagpur — **Kharagpur, West Bengal**
B.Tech. in Manufacturing Science & Engineering + M.Tech. in Industrial & Systems Engineering — *Nov'22 – Jun'27*

Internships

Research Fellow (MARS 4.0) | *Cambridge AI Safety Hub* | *Cambridge, UK (Hybrid)* | *[Link]* — ***Dec'25 - Present***
Under Prof. Fernando Rosas, studying how sequence models represent Bayesian beliefs under coarse-graining and control

- Implemented **input-HMM→transducer** pipelines for exact updates of joint, marginal & coarse-grained Bayesian beliefs
- Decoded fully observed beliefs in 4-layer transformer pilots (held-out R^2 =**0.985–0.997**); entropy-rate gap ≤**0.01 nats**
- Quantified coarse-graining: separability fell from **100%** to **34%** (SNS) / **6%** (Fractal2); shuffle/untrained/seq-CV checks

Data Science Intern | *Axis Bank, Business Intelligence Unit* | *Mumbai, India (On-site)* | *[Link]* — ***May'26 - Jul'26***
Engineered a leak-free temporal graph pipeline over 224.8M accounts, 1.31B transfers, to rank loan onboarding fraud

- Cast fraud proximity as bidirectional 3-hop time-respecting BFS, **pruning** the graph **20×** at **80%** applicant **coverage**
- Hand-rolled Bahmani $(2+2\varepsilon)$ peeling with $1/\log_2(\text{fan-in})$ camouflage weights; HDFS checkpoints truncated RDD lineage
- Shipped a LOAO-validated review queue at 81× top-band lift; WOE-IRLS scorecard confirmed a feature-not-model ceiling

Research Intern | *Complex Networks Research Lab* | *IIT Kharagpur (On-site)* | *[Link] [LoR]* — ***Jul'25 - May'26***
Under Prof. Pawan Goyal, developed LaViDA, a reward-gated latent auxiliary loss augmenting GRPO for a math LLM pipeline

- Revealed RL/SFT tie at **n=8** despite greedy divergence via opposite dynamics: distribution-shaping vs mode-narrowing
- Built VRAM-**optimized GRPO** loops (LoRA-r64, vLLM, flash-attn) scaling 16 rollouts on a single H100 at 36s/step
- Designed **Oracle-augmented** MSE alignment, achieving 79.57% n=8 mean correctness (**+4.70pp** over GRPO, p=.007)

Research Intern | *RAAPID INC* | *Remote* | *[Link] [LoR]* — ***Mar'25 - Jul'26***
Under Prof. Amitava Das, architected GRIT from scratch, optimized its GPU stack, and overhauled a clinical NER pipeline

- Authored **Triton** kernels for fused covariance, GPU **Cholesky** & preconditioning, cutting GMEM traffic and CPU syncs
- Architected **async CUDA** streams overlapping K-FAC inversions/eigensolves with training across **60+** LoRA modules
- Rebuilt clinical NER eval & deterministic QC: **0.640 exact micro-F1**, **zero subword fragments** across all 5,035 rows

Publications

GRIT: Geometry-aware PEFT | *arXiv Preprint* | *[Code]* | *[Paper]* — ***Mar'25 - Present***
Sole first author of "GRIT: Geometry-aware PEFT via rank-space K-FAC, Fisher reprojection & dynamic rank adaptation"

- Constructed a **second-order** PEFT framework utilizing rank-space **natural gradients** to accelerate LLM **convergence**
- Calibrated adaptive **Fisher-spectrum** thresholding to **dynamically** adjust LoRA ranks and **optimize** parameter usage
- Matched LoRA/QLoRA with **25-80%** lower effective rank; exported a **32% smaller adapter** at **<0.04pp** accuracy loss

Memory Transformers | *ICLR 2026 NFAM Workshop* | *[Code]* | *[Paper]* — ***Mar'26***
Co-authored "Mixture of Chapters: Scaling Learnt Memory in Transformers" accepted at ICLR 2026 NFAM Workshop

- Proposed sparse learnable memory banks and cross-attention layers for latent storage and retrieval beyond RAG systems
- Developed MoE-inspired **chapter routing** to partition memory banks, enabling scaling to **262K** learned memory tokens
- Outperformed iso-FLOP vanilla transformers on pretraining and knowledge benchmarks, with lower forgetting during IFT

Competitions

Runner-up | *General Championship Data Analytics, IIT Kharagpur* | *[Link]* — ***Mar'26***
Captained the team to build a full-stack GenAI analytics dashboard for Frammer AI, delivering insights via NLQ and KPI labs

- Orchestrated a NLQ system to drive dashboard insights & KPI analysis (LangGraph, self-healing SQL, 90% EX on BIRD)
- Augmented evaluation capabilities using Gaussian-anchored synthetic data generation to test frameworks on a star schema

Participant | *Amazon ML Challenge 2025* | *[Link]* — ***Oct'25***
Secured 40.8 SMAPE by stacking Qwen2.5-VL-3B SFT & LightGBM on CLIP/text features via a price-space meta-learner

- Eliminated I/O bottlenecks via **offline tensorization** & **WebDataset**, accelerating 4-bit QLoRA throughput by **~75%**
- Enforced alignment via special-token masking & chat templates, using **Pseudo-Huber** loss with monotonic constraints

National Finalist | *Decision Science Track - The American Express Campus Challenge* | *[Link]* — ***Jul'25***
Achieved 0.59 MAP score on the final, unseen evaluation set with a 3-stage GBDT-Transformer ensemble for offer ranking

- Engineered **3k+** features with a **parallelized** pipeline, creating leakage-free profiles with advanced temporal metrics
- Trained a Transformer on **GBDT residuals** using a **listwise** ranking loss to correct the ensemble's systematic errors

Technical Skills

- **Languages/ Tools** : Python, C++, CUDA, SQL, PySpark, Hadoop, Linux, Docker, SLURM, vLLM, LangGraph
- **Libraries :** PyTorch, JAX, Triton, FSDP/DTensor, Transformers, FlashAttention-2, PEFT/LoRA, bitsandbytes, TRL

Key Courses Taken

- **University:** Safety Fundamentals of Generative AI | Operations Research | Probability & Statistics | Linear Algebra
- **MOOCs:** Stanford CS229 (ML) | Stanford CS230 (DL) | LLM Agents MOOC | Algozenith | Summer Analytics

Achievements

Codeforces **Pupil** (Handle: **pritish171**) | Demonstrated Academic Excellence with **95.60%** in Grade 10 (ICSE) and **90.80%** in Grade 12 (CBSE) | Secured merit positions in JEE Main, JEE Advanced, and WBJEE

Extra-Curricular Activities

Represented Patel Hall in Football and Water Polo Teams for Interhall Sports General Championship | Karate black belt with participation in multiple tournaments | NSS volunteer, contributed to improving living conditions in nearby villages